

# A Two-Phase Particle Swarm and Variable Neighborhood Search Algorithm for Continuous Wind Farm Layout Optimization

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

**Abstract**—The wind farm layout optimization problem (WFLOP) places a prescribed number of turbines to minimize wake losses subject to boundary and minimum-spacing constraints. This paper proposes PSO-VNS, a two-phase algorithm for the continuous WFLOP: particle swarm optimization (PSO) with the constriction coefficients of Clerc and Kennedy explores the layout space for the first half of the evaluation budget, and the basic variable neighborhood search (VNS) then refines the global best layout. Both phases minimize one penalized objective based on the Jensen wake model and Weibull wind data. On 68 benchmark cases with 30 seed-paired runs of 6,030 evaluations, PSO-VNS returns a feasible layout in every run and obtains the best average rank. It is at least as good as a correctly configured PSO and better for larger numbers of turbines (14 significant wins and 4 losses; 9 wins and 2 losses for 10 or more turbines), and it clearly outperforms a salp swarm variant of the same design (SSA-VNS; 49 wins, no loss), the salp swarm algorithm (SSA), the Laplacian SSA, differential evolution, VNS and multistart SLSQP. The ablation shows that salp swarms are weaker first phases than a convergent PSO and in most cases tied with random sampling, and that the Laplace step gives no gain. We also find that the common PSO setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  lies outside the convergence region of PSO and makes PSO an overly weak baseline. On a 16-turbine block of Horns Rev 1, the best PSO-VNS layout exceeds the energy production of the installed layout. With up to 20 times the budget, PSO-VNS keeps the best average rank at all three budgets; on IEA Wind Task 37 Case Study 1, the best PSO-VNS layouts at 30,030 evaluations reach 98.7% (16 turbines) and 95.8% (36 turbines) of the AEP of the best feasible published layouts.

**Index Terms**—Wind farm layout optimization, particle swarm optimization, constriction coefficient, variable neighborhood search, hybrid metaheuristic, wake effect.

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## NOMENCLATURE

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| $B$                        | Budget (number of objective calls).                |
| $c_1, c_2$                 | Acceleration coefficients of PSO.                  |
| $C_T$                      | Thrust coefficient.                                |
| $D, R$                     | Rotor diameter and rotor radius ( $D = 2R$ ).      |
| $d_{ij}$                   | Distance of turbine $i$ behind turbine $j$ .       |
| $\Delta_{ij}, \Delta_i$    | Deficit at $i$ due to $j$ ; total deficit at $i$ . |
| $E(P_i)$                   | Expected power of turbine $i$ (benchmark units).   |
| $E_{ideal}$                | Wake-free expected power of one turbine.           |
| $F_p$                      | Penalized wake loss (minimized).                   |
| $g_i^b, g_{ij}^s$          | Boundary and spacing violations.                   |
| $\mathbf{G}, \mathbf{P}^i$ | Global best; personal best of particle $i$ .       |
| $\mathbf{H}$               | Food source (best layout of SSA).                  |
| $K$                        | Jensen wake-spreading constant.                    |
| $k, k_{\max}$              | Index and number of VNS neighborhoods.             |
| $\ell_{ij}, \ell_{\min}$   | Distance of $i$ and $j$ ; minimum spacing.         |
| $N, N_p$                   | Number of turbines; population size.               |
| $N_\theta, N_s$            | Numbers of wind-direction and wind-speed bins.     |
| $p_l$                      | Probability of the $l$ th wind-direction bin.      |
| $r$                        | Farm radius.                                       |
| $s$                        | Wind speed.                                        |
| $t, T, T_1$                | Iteration; total and Phase-1 iterations.           |
| $\mathbf{V}^i$             | Velocity of particle $i$ .                         |
| $w$                        | Inertia weight of PSO.                             |
| $\mathbf{X}$               | Layout vector $(x_1, y_1, \dots, x_N, y_N)$ .      |
| $\alpha, \beta_{ij}$       | Half-cone angle; wake angle of $i$ behind $j$ .    |
| $\zeta, \psi$              | Weibull shape and scale parameters.                |
| $\theta$                   | Wind direction (blowing towards, CCW from east).   |
| $\kappa, \varphi_{1,2}$    | Constriction factor and coefficients.              |
| $\mu$                      | Penalty factor.                                    |
| $\omega$                   | Fraction of the budget given to Phase 1.           |

## I. INTRODUCTION

Wind energy has become one of the main sources of low-carbon electricity, and the placement of the turbines, fixed at the planning stage, has a lasting effect on the energy a farm delivers. The wind farm layout optimization problem (WFLOP) addresses this decision. An upstream turbine leaves behind a wake of slower and more turbulent air that reduces the power of the turbines downstream; the wake recovers with distance, but the site is limited and the turbines must keep a minimum distance from each other. The objective is nonlinear, the wakes couple the positions of all turbines, and the feasible

region is bounded by many nonconvex spacing constraints, which has made WFLOP a popular test bed for metaheuristics.

Early studies placed turbines on a grid with genetic algorithms (GAs) [1], [2]. Kusiak and Song [3] introduced the continuous, wind-distribution-based model that we also use, and it has since been solved with ant colony optimization and particle filtering [4], [5] and with biogeography-based optimization [6]. GA, particle swarm optimization (PSO), differential evolution (DE) and simulated annealing were compared in [7], PSO has been applied to reduce wake losses [8], and the collector and electrical systems of offshore farms have been optimized and reviewed in [9], [10]. Later work used informed GAs [11], [12], deterministic refinement [13], variable neighborhood search (VNS) for large offshore farms [14], pseudo-gradients [15], Gaussian wake models [16] and surrogates [17]–[19]. The IEA Wind Task 37 case study [20] showed that, even with a common wake model, different optimizers reach clearly different layouts.

Many of these studies propose a new or modified metaheuristic and compare it with reference methods, among which PSO is one of the most common. Such comparisons are only as informative as the configuration of the references. The particles of PSO converge only if its inertia weight  $w$  and acceleration coefficients  $c_1$ ,  $c_2$  lie in a known region [21], [22], and the setting  $w = 0.7$ ,  $c_1 = c_2 = 2$ , which is common in PSO baselines and was used in the earlier version of this study, lies outside it. With the constriction coefficients of Clerc and Kennedy [21], plain PSO outperforms the salp swarm algorithm (SSA) [23], the Laplacian SSA (LX-SSA) [24] proposed earlier by two of the authors, and an SSA-VNS hybrid that we had developed first (Section VI). A baseline outside its convergence region can therefore make a new method look stronger than it is; we report this as a finding about the configuration of baselines, not about any particular study.

Swarms and trajectory methods complement each other. Once a swarm has contracted around its best layout, the remaining evaluations bring small gains, because the final refinement requires small moves of individual turbines along the active spacing constraints, whereas VNS [25], [26] refines one layout efficiently but depends on where it starts. A swarm-then-VNS design helps only if the swarm provides a better start than cheaper alternatives, and SSA did not: SSA-VNS was statistically tied with a control that spends the same evaluations on random layouts (Section VI-D). A convergent PSO, which contracts around a good region within its share of the budget, is the natural alternative.

We therefore propose PSO-VNS: PSO with constriction coefficients explores the layout space for the first half of the budget, and the basic VNS refines the global best layout of the swarm. Both components are standard, and the gain over a well-configured PSO is modest; the contribution lies in their combination for the constrained WFLOP, in a controlled evaluation of which ingredients matter, and in the finding on the configuration of PSO baselines. We keep the Jensen wake model, Weibull wind data and the Kusiak–Song power model for comparability with the literature and examine these choices separately. Our contributions are as follows:

- 1) We propose PSO-VNS, a two-phase algorithm for the continuous WFLOP with explicit boundary and minimum-spacing constraints, in which a convergent PSO and the basic VNS share one penalized objective (Section IV).
- 2) We compare PSO-VNS with PSO, SSA-VNS, SSA, LX-SSA, DE, VNS and a gradient-based multistart SLSQP solver on 68 benchmark cases at equal numbers of objective evaluations with 30 seed-paired runs, and isolate in an ablation the VNS phase, the value of the swarm start (against VNS alone and a random-multistart control), the choice of the first-phase swarm and the Laplace step of LX-SSA (Section VI).
- 3) We show that the PSO setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  violates the convergence condition of PSO and quantify how much it understates PSO (Sections III-A and VI-C).
- 4) We test the methods with feasibility-preserving initialization, budgets of up to 120,030 evaluations, a 16-turbine block of the Horns Rev 1 offshore farm, the 16- and 36-turbine scenarios of IEA Wind Task 37 Case Study 1 (IEA37), and alternative power-curve, wake, spacing and boundary assumptions (Sections VII and VIII).

Section II describes the model, Section III the swarm methods, and Section IV PSO-VNS. Sections V–VIII give the setup and results, Section IX the limitations, and Section X concludes the paper.

## II. PROBLEM MODELING

The number of turbines  $N$  is prescribed in every run, and only the  $2N$  coordinates  $(x_i, y_i)$  of the identical turbines, measured from the center of the farm, are optimized; results for several  $N$  form a sweep. For a wind direction  $\theta$ , the wind speed  $s$  follows a Weibull distribution with density  $p_s(s; \zeta, \psi) = (\zeta/\psi)(s/\psi)^{\zeta-1} \exp[-(s/\psi)^\zeta]$ , whose shape  $\zeta(\theta)$  and scale  $\psi(\theta)$  depend on the direction. Throughout the paper,  $\theta$  is the direction *towards* which the wind blows, counter-clockwise from east; meteorological directions  $\theta_{\text{met}}$  (wind *from*, clockwise from north), as given for Horns Rev 1 and IEA37, are converted by  $\theta = 270^\circ - \theta_{\text{met}}$ . Any two turbines must be at least  $\ell_{\min} = 4D = 8R$  apart,  $\ell_{ij} = \|(x_i - x_j, y_i - y_j)\|_2 \geq \ell_{\min}$ , as in the benchmark studies [3], [6] (Section VIII-B discusses  $5D$  and  $6D$ ), and the farm is circular,  $x_i^2 + y_i^2 \leq r^2$ , with  $r = 500, 750$  and  $1000$  m. We use the Jensen wake model [27], [28] and the power model of [3], and maximize the expected power of the farm, i.e., minimize its wake loss, subject to both constraints.

### A. Wake Effect Model

In the Jensen model [27], [28], the wake behind a rotor expands linearly with the downstream distance. With the wake-spreading constant  $K$ , the thrust coefficient  $C_T$  and the distance  $d_{ij}$  of turbine  $i$  behind turbine  $j$ , the velocity deficit is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \quad (1)$$

The wake of turbine  $j$  is a half cone with half-angle  $\alpha = \arctan K$  whose virtual vertex lies  $R/K$  upstream of the rotor. Turbine  $i$  lies in it if the wake angle  $\beta_{ij} = \cos^{-1}(u_{ij}/\sqrt{u_{ij}^2 + v_{ij}^2})$  satisfies  $\beta_{ij} < \alpha$ , where  $u_{ij} = (x_i - x_j) \cos \theta + (y_i - y_j) \sin \theta + R/K$ , the lateral offset from the wake axis is  $v_{ij} = -(x_i - x_j) \sin \theta + (y_i - y_j) \cos \theta$ , and  $d_{ij} = |u_{ij} - R/K|$ . The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left( \sum_{j \neq i, \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \quad (2)$$

Within a direction bin,  $\Delta_i(\theta)$  reduces the free-stream speed by a constant factor. Since  $aS \sim \text{Weibull}(\zeta, a\psi)$  for  $a > 0$  when  $S \sim \text{Weibull}(\zeta, \psi)$ , the local wind speed keeps the shape parameter and has the scale parameter

$$\psi_i(\theta) = \psi(\theta) [1 - \Delta_i(\theta)], \quad i = 1, \dots, N. \quad (3)$$

Schematics of the farm and the wind-direction convention, the wake model and the wake cone are given in Figs. S1–S3 of the supplementary material.

### B. Power Model

We use the power curve of the Kusiak–Song benchmark [3]:  $f(s) = 0$  for  $s < s_{\text{cut-in}}$ ,  $f(s) = \lambda_1 s + \lambda_0$  for  $s_{\text{cut-in}} \leq s \leq s_{\text{rated}}$  and  $f(s) = P_{\text{rated}}$  above, with  $s_{\text{cut-in}} = 3.5$  m/s,  $s_{\text{rated}} = 14$  m/s,  $P_{\text{rated}} = 1500$  kW,  $\lambda_1 = 140.86$  and  $\lambda_0 = -500$ . We use this curve exactly as published, although its linear part is slightly negative just above the cut-in speed and reaches 1472 kW at the rated speed, where it jumps to 1500 kW; the benchmark has no cut-out speed. The expected power of turbine  $i$ ,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s; \zeta(\theta), \psi_i(\theta)) ds d\theta, \quad (4)$$

is evaluated, as in [3], by a Riemann sum over  $N_\theta$  direction bins with probabilities  $p_l$  and  $N_s$  speed bins between  $s_{\text{cut-in}}$  and  $s_{\text{rated}}$  (Section S-I.A of the supplementary material). Like the earlier studies, we keep the bin width in degrees as a factor of the direction sum. Because all 24 direction bins are  $15^\circ$  wide and  $\sum_l p_l = 1$ , the benchmark objective  $\sum_i E(P_i)$  equals 15 times the expected farm power in kW, and  $\text{AEP [MWh]} = (\text{objective}/15) \times 8.76$ ; e.g., the wake-free benchmark objective of one turbine under Wind Data Set I,  $E_{\text{ideal}} = 14045.74$ , corresponds to 936.38 kW or 8202.7 MWh per year.

### C. Constraint Handling

After every position update, each coordinate is clipped to the square that encloses the farm,  $x_i \leftarrow \min\{\max\{x_i, -r\}, r\}$  and likewise for  $y_i$  (box repair). The circular boundary and the spacing are handled by a penalty. With the violations  $g_{ij}^s = \ell_{\min} - \ell_{ij}$  and  $g_i^b = x_i^2 + y_i^2 - r^2$ , a layout is feasible if  $g_{ij}^s \leq 0$  and  $g_i^b \leq 0$  for all  $i$  and  $j$ . All penalty-based optimizers minimize the penalized wake loss

$$F_p(\mathbf{X}) = \left( NE_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \mu g_i^b)^2 + \sum_{i < j: g_{ij}^s > 0} (1 + \mu g_{ij}^s)^2, \quad \mu = 10^{10}, \quad (5)$$

where  $\mathbf{X} = (x_1, y_1, \dots, x_N, y_N)$  and  $NE_{\text{ideal}}$  is the wake-free benchmark objective, so that the first term is the wake loss reported in our tables. Any violation larger than about  $10^{-7}$  (m or  $\text{m}^2$ ) makes the penalty larger than the largest possible wake loss. Comparing  $F_p$  values therefore gives a feasibility-first selection: a feasible layout is always preferred to an infeasible one, two feasible layouts are compared by their wake loss, and two infeasible layouts by their total penalty.

## III. PRELIMINARIES: SWARM METHODS FOR PHASE I

All swarm methods below keep a population of  $N_p$  candidate vectors  $\mathbf{X}^i \in \mathbb{R}^n$  with components  $X_m^i$ ,  $m = 1, \dots, n$ , bounds  $lb_m$ ,  $ub_m$ , and cost  $N_p$  objective calls per iteration unless stated otherwise.

### A. Particle Swarm Optimization with Constriction Coefficients

In PSO [29], particle  $i$  has a position  $\mathbf{X}^i$ , a velocity  $\mathbf{V}^i$  and a personal best  $\mathbf{P}^i$ , and the swarm keeps the global best  $\mathbf{G}$ . In the inertia-weight form [30], a particle moves by

$$\mathbf{V}^i \leftarrow w \mathbf{V}^i + c_1 \boldsymbol{\rho}_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2 \boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i), \quad (6)$$

$$\mathbf{X}^i \leftarrow \mathbf{X}^i + \mathbf{V}^i, \quad (7)$$

where  $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2$  are vectors of independent  $U(0, 1)$  numbers, drawn anew for every move, and  $\circ$  is the componentwise product. Clerc and Kennedy [21] multiply the bracket  $\mathbf{V}^i + \varphi_1 \boldsymbol{\rho}_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + \varphi_2 \boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i)$  by the constriction factor  $\kappa = 2 / |2 - \varphi - \sqrt{\varphi^2 - 4\varphi}|$ ,  $\varphi = \varphi_1 + \varphi_2$ . The standard choice  $\varphi_1 = \varphi_2 = 2.05$  gives  $\kappa = 0.7298$ , which is (6) with

$$w = 0.7298, \quad c_1 = c_2 = \kappa \varphi_1 = 1.49618. \quad (8)$$

When the personal and global bests stay fixed (stagnation), the mean and the variance of the particle positions converge if  $|w| < 1$  and [22]

$$c_1 + c_2 < \frac{24(1 - w^2)}{7 - 5w}. \quad (9)$$

For the constriction setting (8), the right-hand side equals 3.35 and  $c_1 + c_2 = 2.99$  satisfies (9). For  $w = 0.7$  the bound is 3.50, so the common setting  $c_1 = c_2 = 2$  ( $c_1 + c_2 = 4$ ) violates it: the variance of the positions then grows even when the bests no longer change, and the swarm keeps sampling far from its best layouts instead of refining them. In our implementation, the velocities start at zero and are not bounded, the positions are box-repaired after each move, and  $\mathbf{P}^i$  and  $\mathbf{G}$  are updated immediately after each particle is evaluated.

### B. Salp Swarm Algorithm and Its Laplacian Variant

The SSA [23] sorts the population by the objective, keeps the best solution found so far as the food source  $\mathbf{H}$ , and splits the population into a leader half and a follower half. A leader  $i \leq N_p/2$  updates each coordinate around the food source,

$$X_m^i = H_m \pm \xi_1 ((ub_m - lb_m)\xi_2 + lb_m), \quad (10)$$

with the sign chosen with probability 1/2 by  $\xi_3$ , where  $\xi_2, \xi_3 \sim U(0, 1)$  and  $\xi_1 = 2 \exp[-(4t/T)^2]$  decreases with the iteration  $t$  out of  $T$ . A follower  $i > N_p/2$  moves to

the midpoint between itself and the preceding salp,  $X_m^i \leftarrow \frac{1}{2}(X_m^i + X_m^{i-1})$ . LX-SSA [24] gives each follower a second, Laplace-based candidate [31]

$$Y_m^i = X_m^i + \gamma (H_m - X_m^i),$$

$$\gamma = \phi - \chi \operatorname{sgn}(z - \frac{1}{2}) \ln(1 - 2|z - \frac{1}{2}|), \quad (11)$$

with  $z \sim U(0, 1)$ , i.e., a Laplace variable with location  $\phi = 0$  and scale  $\chi = 1$ , and the follower keeps the better of its two candidates. One LX-SSA iteration therefore costs  $2N_p$  calls. The comparison of SSA and LX-SSA isolates the Laplace step. The pseudocode of both methods is given in Section S-II of the supplementary material.

#### IV. PROPOSED PSO-VNS ALGORITHM

PSO-VNS combines a swarm that explores the layout space and settles around a good region with a trajectory method that refines one layout.

##### A. Application to the WFLOP

For a prescribed turbine count  $N$ , a candidate is the layout vector  $\mathbf{X} \in \mathbb{R}^n$ ,  $n = 2N$ , with bounds  $[-r, r]^n$ . Every candidate is box-repaired, its deficits  $\Delta_i(\theta)$  and scale parameters  $\psi_i(\theta)$  are computed for every direction bin by (2) and (3), and  $F_p$  is evaluated by (5). Both phases use  $F_p$  in every comparison, best update and acceptance step, so the feasibility-first rules of Section II-C hold throughout; in particular,  $\mathbf{P}^i$  and  $\mathbf{G}$  move towards feasible layouts as soon as the swarm finds them.

##### B. Phase 1: Exploration by Constriction PSO

For a budget of  $B$  objective calls and a split fraction  $\omega \in (0, 1)$ , PSO with the coefficients (8) runs with population  $N_p$  for

$$T_1 = \operatorname{round}\left(\frac{\omega B - N_p}{N_p}\right)$$

iterations, since the initial swarm costs  $N_p$  calls and every iteration another  $N_p$ . Because the constriction PSO has no iteration-dependent parameter, Phase 1 of PSO-VNS coincides, for a given seed, with the first  $N_p(T_1 + 1)$  calls of a PSO run with the same budget; PSO-VNS and PSO differ only in how they spend the rest of the budget. The global best  $\mathbf{G}$  at the end of Phase 1 is the starting point of Phase 2.

##### C. Phase 2: Refinement by Basic VNS

Phase 2 is the basic VNS of [25], [26] in the continuous form of [32], applied to  $F_p$ . It uses  $k_{\max} = 5$  nested  $\ell_\infty$  neighborhoods of the whole layout,

$$\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k\}, \quad \delta_k = 0.1 k r.$$

In the shaking step, a point is drawn uniformly from  $\mathcal{N}_k(\mathbf{X})$ , so that all  $2N$  coordinates move, and box repair is applied. The shaken layout is then improved by a complete best-improvement local search. Since the top-hat Jensen wake makes  $F_p$  discontinuous, we use a derivative-free compass search: it evaluates all  $2n = 4N$  moves  $\pm h \mathbf{e}_m$  along the

#### Algorithm 1 PSO-VNS for a prescribed turbine count $N$

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**Require:** budget  $B$ , split  $\omega$ , population  $N_p$ ,  $w$ ,  $c_1$ ,  $c_2$ , farm radius  $r$ ,  $k_{\max}$

- 1:  $T_1 \leftarrow \operatorname{round}((\omega B - N_p)/N_p)$  ▷ Phase 1
- 2: Draw and evaluate  $N_p$  particles  $\mathbf{X}^i$ ;  $\mathbf{V}^i \leftarrow \mathbf{0}$ ;  $\mathbf{P}^i \leftarrow \mathbf{X}^i$ ;  $\mathbf{G} \leftarrow \arg \min_i F_p(\mathbf{P}^i)$
- 3: **for**  $t = 1$  to  $T_1$  **do**
- 4:   **for** particle  $i = 1, \dots, N_p$  **do**
- 5:     Update  $\mathbf{V}^i$ ,  $\mathbf{X}^i$  by (6), (7); box repair
- 6:     Evaluate  $F_p(\mathbf{X}^i)$ ; update  $\mathbf{P}^i$  and  $\mathbf{G}$
- 7:   **end for**
- 8: **end for**
- 9:  $\mathbf{X} \leftarrow \operatorname{LOCALSEARCH}(\mathbf{G})$ ;  $k \leftarrow 1$  ▷ Phase 2; compass search
- 10: **while** fewer than  $B$  objective calls have been used **do**
- 11:    $\mathbf{X}' \leftarrow$  uniform random point of  $\mathcal{N}_k(\mathbf{X})$ ; box repair
- 12:    $\mathbf{X}' \leftarrow \operatorname{LOCALSEARCH}(\mathbf{X}')$
- 13:   **if**  $F_p(\mathbf{X}') < F_p(\mathbf{X})$  **then**
- 14:      $\mathbf{X} \leftarrow \mathbf{X}'$ ;  $k \leftarrow 1$
- 15:   **else**
- 16:      $k \leftarrow k \bmod k_{\max} + 1$
- 17:   **end if**
- 18: **end while**
- 19: **return** best layout evaluated

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coordinate axes, moves to the best improving one, and halves the step  $h$  (starting from  $0.05r$ ) when no move improves, until  $h < 10^{-3}r$ . If the resulting local minimum is better than the incumbent, it is accepted and  $k$  is reset to 1; otherwise  $k$  is increased by one and returns to 1 after  $k_{\max}$ . Phase 2 first applies the local search to  $\mathbf{G}$  and then repeats shaking, local search and neighborhood change until the budget  $B$  is used up (Algorithm 1).

##### D. Parameters and Cost

PSO-VNS uses  $N_p = 30$ , the coefficients (8) and the VNS settings above, and it adds a single parameter, the split  $\omega$ . With the default  $\omega = 0.5$  and  $B = 6,030$ , PSO runs for  $T_1 = 100$  iterations and receives  $30 + 100 \times 30 = 3,030$  calls, and VNS the remaining 3,000. None of these values was tuned; the effect of  $\omega$  is studied in Section VI-E. The cost of both phases is dominated by the objective evaluations, so for a budget of  $B$  calls PSO-VNS has the same  $\mathcal{O}(B N^2 N_\theta N_s)$  cost as the other methods; the PSO update, box repair, shaking and each compass move are  $\mathcal{O}(N)$  and the spacing check  $\mathcal{O}(N^2)$ . Because the random-number generator is seeded once and Phase 1 draws the initial population first, PSO-VNS starts from the same initial population as all other methods for a given seed, which keeps the comparisons paired.

#### V. EXPERIMENTAL SETUP

##### A. Benchmark Cases

We use the two wind data sets of the Kusiak–Song benchmark [3], [6], listed in Table S21 of the supplementary material. In Wind Data Set I, the Weibull parameters are the same in all directions ( $\zeta = 2$ ,  $\psi = 13$ ) and the wind

is strongly directional (probabilities 0.2 and 0.6 for the bins  $[75^\circ, 90^\circ)$  and  $[90^\circ, 105^\circ)$ , 0.01 for most others); Wind Data Set II keeps  $\zeta = 2$ , but  $\psi$  varies from 2.6 to 10 m/s with the direction and the wind is spread over more directions. We use  $N_\theta = 24$  direction bins of  $15^\circ$  and  $N_s = 21$  speed intervals of 0.5 m/s. Each data set is combined with three circular farms of radius  $r = 500, 750$  and  $1000$  m, with  $N = 2, \dots, 10, 2, \dots, 12$  and  $2, \dots, 15$  turbines, respectively, which gives  $2 \times (9 + 11 + 14) = 68$  cases. The turbines have  $R = 38.5$  m,  $\ell_{\min} = 4D = 308$  m,  $C_T = 0.8$  and  $K = 0.075$ .

### B. Methods Compared

The main comparison includes PSO-VNS and seven reference methods, all evaluated with the objective (5), the bounds  $[-r, r]^{2N}$ , population 30 where applicable, and the same budget:

- **PSO-VNS** (proposed): Algorithm 1 with  $\omega = 0.5$ , i.e., 100 PSO iterations (3,030 calls) followed by 3,000 calls of VNS.
- **PSO** [21], [29]: the coefficients (8) and 200 iterations. The setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  of the earlier version of this study is discussed in Section VI-C.
- **SSA-VNS**: Algorithm 1 with SSA in Phase 1 (100 SSA iterations, 3,030 calls,  $\xi_1$  decreasing over these 100 iterations).
- **SSA** [23] (200 iterations), **LX-SSA** [24] ( $\phi = 0$ ,  $\chi = 1$ , 100 iterations) and **DE** [33] (DE/rand/1/bin, scale factor 0.5, crossover rate 0.9, 200 iterations).
- **VNS**: Phase 2 of Algorithm 1, started from the best member of the initial population.
- **Multistart SLSQP (MS-SLSQP)**: SciPy's SLSQP [34] with explicit spacing and boundary constraints and forward-difference gradients (1-m step), started from the initial population in the order of the  $F_p$  values and restarted until the budget is used up.

The ablation adds two variants. **RS-VNS** is a random-multistart control: Phase 1 is replaced by pure random sampling with the same number of calls ( $\text{round}(\omega B) = 3,015$  layouts, including the initial population), and the best sampled layout is refined by the same VNS. **LX-SSA-VNS** uses LX-SSA in Phase 1 ( $T_1 = \text{round}((\omega B - N_p)/(2N_p)) = 50$  iterations, 3,030 calls). SSA, LX-SSA, PSO and DE use our original implementation from the earlier version of this study, with the coefficients (8) for PSO; VNS, RS-VNS, MS-SLSQP and the three hybrids were implemented for this study, and the hybrids call the original swarm code in Phase 1.

Every run of the main comparison has a budget of exactly 6,030 objective calls, counted by a wrapper around the objective function (including the initial population and, for MS-SLSQP, the finite-difference gradient calls). Each method is run with the seeds 1–30; every optimizer seeds the random-number generator and draws its initial population first, so runs with the same seed start from the *same* initial population for all methods and are paired.

### C. Tests Beyond the Benchmark

**Initialization and budget:** In the main comparison, the initial layouts are drawn uniformly in the bounding square and

are usually infeasible. As an alternative, each initial layout is drawn inside the site and pushed apart by minimizing a smooth packing penalty with L-BFGS-B until the constraints hold, without objective calls. The nine methods other than RS-VNS are run with this initialization at 6,030 calls (for RS-VNS, feasible sampling would require one packing solve for each of its 3,015 random layouts per run, which is far more expensive than the objective calls themselves), and all ten methods (the eight of the main comparison, RS-VNS and LX-SSA-VNS) are run with random initialization at 6,030, 30,030 and 120,030 calls, on the six largest cases (the largest  $N$  of each farm and data set) and the Horns Rev 1 block, with 30 seeds (10 for Horns Rev 1 at 120,030 calls). All iteration counts scale with the budget; e.g., PSO-VNS uses 500 PSO iterations at 30,030 calls.

**Horns Rev 1 16-turbine block:** We use the Horns Rev 1 offshore site from PyWake [35]: Vestas V80 turbines (rotor diameter 80 m, 2 MW) with tabulated power and thrust curves and a 25 m/s cut-out, the measured 12-sector wind climate, the parallelogram through the corner turbines of the block as boundary, the Jensen wake with  $K = 0.04$  (AEP within 0.3% of PyWake's), and PyWake's 16-turbine example block with  $4D$  (320 m) spacing.

**IEA37 Case Study 1:** Case Study 1 of IEA Wind Task 37 [20], [36] fixes the wake model and leaves the optimizer free; we use its 16- and 36-turbine scenarios (circular boundaries of radius 1300 and 2000 m, minimum spacing  $2D = 260$  m, IEA 3.35-MW reference turbine with  $D = 130$  m, 16 wind directions with a constant speed of 9.8 m/s, simplified Bastankhah Gaussian wake). Our implementation reproduces the official AEP calculator to a relative deviation below  $10^{-11}$ . All methods are run with the penalty (5) applied to the AEP loss, at 6,030 and 30,030 calls with 30 seeds, and compared with the submitted layouts [20], [36].

### D. Performance Measures and Statistical Analysis

For each run we record the final benchmark objective, the wake loss (as a percentage of the wake-free objective), whether the final layout is feasible (spacing and boundary satisfied within  $10^{-6}$  m), the convergence curve (best feasible objective at 201 equally spaced checkpoints), and the wall-clock time. Means and standard deviations are computed over the feasible runs. Within each case, the 30 seed-paired runs are compared by two-sided Wilcoxon signed-rank tests of PSO-VNS against each other method, with Holm's correction [37] over the comparisons of the case and matched-pairs rank-biserial correlations  $r_{\text{rb}}$  as effect sizes; a feasible run beats an infeasible one. Across the 68 cases, the methods are ranked within each case by the mean benchmark objective of their feasible runs and compared by a Friedman test with the Iman–Davenport correction and Holm-adjusted post hoc tests on the average ranks [38]. Because mean-rank post hoc tests depend on the set of compared methods [39], we also compare PSO-VNS with each method by Wilcoxon signed-rank tests on the per-case mean wake losses.

TABLE I

CASE-LEVEL ANALYSIS OVER THE 68 BENCHMARK CASES. AVG. RANK: AVERAGE RANK OF THE MEAN FEASIBLE OBJECTIVE (1 = BEST; METHODS WITH FEWER THAN 15 FEASIBLE RUNS IN A CASE RANKED LAST); “BEST”: CASES IN WHICH THE METHOD ALONE RANKS FIRST; “FEAS.”: FEASIBLE RUNS (%);  $p_z$ : HOLM-ADJUSTED  $p$  OF THE AVERAGE-RANK TEST AGAINST PSO-VNS. BECAUSE MEAN-RANK TESTS DEPEND ON THE POOL OF METHODS [39],  $p_W$  AND  $\Delta L$  GIVE THE HOLM-ADJUSTED TWO-SIDED WILCOXON SIGNED-RANK TEST ON THE 68 PER-CASE MEAN WAKE LOSSES AND THE MEAN WAKE-LOSS DIFFERENCE (PERCENTAGE POINTS, PSO-VNS MINUS METHOD; NEGATIVE = PSO-VNS BETTER)

| Method   | Avg. rank | Best | Feas. | $p_z$                 | $p_W$                 | $\Delta L$ |
|----------|-----------|------|-------|-----------------------|-----------------------|------------|
| PSO-VNS  | 1.82      | 29   | 100.0 | –                     | –                     | –          |
| PSO      | 1.97      | 25   | 100.0 | 0.713                 | 0.296                 | –0.018     |
| SSA-VNS  | 3.45      | 0    | 99.6  | $2.0 \times 10^{-4}$  | $5.7 \times 10^{-11}$ | –0.330     |
| VNS      | 4.40      | 3    | 100.0 | $2.4 \times 10^{-9}$  | $1.3 \times 10^{-10}$ | –0.465     |
| SSA      | 5.38      | 0    | 97.6  | $8.3 \times 10^{-17}$ | $4.5 \times 10^{-11}$ | –0.611     |
| MS-SLSQP | 5.88      | 1    | 100.0 | $1.8 \times 10^{-21}$ | $4.2 \times 10^{-11}$ | –0.435     |
| LX-SSA   | 6.06      | 0    | 96.9  | $3.3 \times 10^{-23}$ | $4.5 \times 10^{-11}$ | –0.765     |
| DE       | 7.04      | 0    | 71.4  | $1.0 \times 10^{-34}$ | $5.6 \times 10^{-11}$ | –0.887     |

Friedman  $\chi^2_F = 319.3$  (7 d.f.),  $p = 4.6 \times 10^{-65}$ ; Iman–Davenport  $F_F = 136.5$

TABLE II

PAIRWISE OUTCOME OF PSO-VNS AGAINST EACH METHOD: CASES IN WHICH PSO-VNS IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (TWO-SIDED WILCOXON SIGNED-RANK TEST, 30 SEED-PAIRED RUNS, HOLM-ADJUSTED OVER THE SEVEN COMPARISONS OF EACH CASE,  $\alpha = 0.05$ ; AN INFEASIBLE RUN RANKS BELOW EVERY FEASIBLE RUN); LAST ROW: MEDIAN RANK-BISERIAL CORRELATION (POSITIVE = PSO-VNS BETTER)

| DS               | $r$ (m) | Cases | PSO     | SSA-VNS | SSA    | LX-SSA | DE      | VNS     | MS-SLSQP |
|------------------|---------|-------|---------|---------|--------|--------|---------|---------|----------|
| I                | 500     | 9     | 0/8/1   | 6/3/0   | 8/1/0  | 8/1/0  | 7/2/0   | 6/3/0   | 7/2/0    |
| I                | 750     | 11    | 1/8/2   | 9/2/0   | 10/1/0 | 10/1/0 | 9/2/0   | 8/3/0   | 10/1/0   |
| I                | 1000    | 14    | 3/11/0  | 9/5/0   | 12/2/0 | 12/2/0 | 12/2/0  | 12/2/0  | 12/2/0   |
| II               | 500     | 9     | 3/5/1   | 7/2/0   | 7/2/0  | 8/1/0  | 7/2/0   | 7/1/1   | 8/1/0    |
| II               | 750     | 11    | 2/9/0   | 8/3/0   | 10/1/0 | 10/1/0 | 9/2/0   | 8/3/0   | 9/2/0    |
| II               | 1000    | 14    | 5/9/0   | 10/4/0  | 12/2/0 | 12/2/0 | 12/2/0  | 10/4/0  | 13/1/0   |
| All              |         | 68    | 14/50/4 | 49/19/0 | 59/9/0 | 60/8/0 | 56/12/0 | 51/16/1 | 59/9/0   |
| $\tilde{r}_{rb}$ |         |       | +0.08   | +0.72   | +0.98  | +0.97  | +1.00   | +0.84   | +0.99    |

### E. Implementation Verification and Environment

Our vectorized objective agrees with the original implementation within a relative difference of  $10^{-6}$  and reproduces the objective values of 720 archived final layouts within  $8.7 \times 10^{-11}$ ; re-running the original SSA, LX-SSA, PSO and DE code with the archived settings reproduces the archived result distributions (22 of 24 Mann–Whitney comparisons with  $p \geq 0.05$ ). All new runs used Linux containers (Intel Xeon processor at 2.1 GHz, 4 virtual cores) with Python 3.11.15, NumPy 2.4.6 and SciPy 1.17.1.

## VI. RESULTS ON THE BENCHMARK

This section reports the 68 benchmark cases (20,400 runs) and the ablation. Per-case means and standard deviations are given in Tables S1–S6 of the supplementary material.

### A. Overall Comparison

Table I and Fig. 1 summarize the comparison over all cases. The Friedman test rejects the hypothesis that the eight methods perform equally ( $\chi^2_F = 319.3$ ,  $p = 4.6 \times 10^{-65}$ ). PSO-VNS has the best average rank (1.82) and alone ranks first

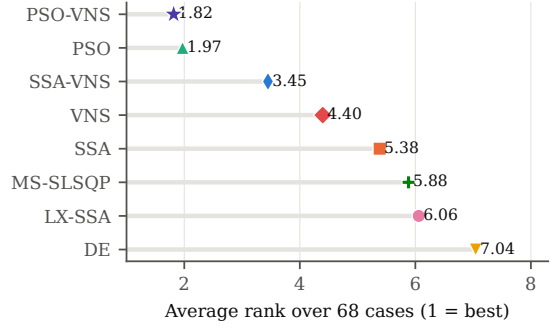


Fig. 1. Average rank of the eight methods over the 68 benchmark cases (1 = best; ranking rule of Table I).

in 29 cases, followed by PSO (1.97), SSA-VNS (3.45), VNS (4.40), SSA (5.38), MS-SLSQP (5.88), LX-SSA (6.06) and DE (7.04). In the Holm-adjusted post hoc tests, PSO-VNS ranks significantly better than SSA-VNS, VNS, SSA, MS-SLSQP, LX-SSA and DE ( $p_{\text{Holm}} \leq 2.0 \times 10^{-4}$ ), but not than PSO ( $p_{\text{Holm}} = 0.71$ ).

The run-level tests in Table II show where these differences come from. PSO, which forms the first phase of PSO-VNS, is the closest competitor: PSO-VNS is significantly better in 14 cases, worse in 4 and tied in 50, and its mean wake loss over the 68 cases is 1.322% against 1.341%. Over all cases this difference is not significant (two-sided Wilcoxon signed-rank test on the 68 per-case mean wake losses,  $p = 0.30$ ), but it concentrates in the larger cases: for  $N \geq 10$ , PSO-VNS wins 9 of the 20 cases and loses 2, with a mean gain of 0.08 percentage points, whereas for  $N < 10$  both methods reach nearly the same wake loss (mean absolute difference 0.03 percentage points). PSO-VNS is thus at least as good as a correctly configured PSO and better for larger numbers of turbines. Against SSA-VNS, the same design with a salp swarm first phase, PSO-VNS is significantly better in 49 cases and never worse (1.322% vs. 1.653%,  $p = 1.9 \times 10^{-11}$ ). Against SSA, LX-SSA and DE, it is significantly better in 59, 60 and 56 cases, respectively, and against VNS and MS-SLSQP in 51 and 59 cases; it is never significantly worse than SSA-VNS, SSA, LX-SSA, DE and MS-SLSQP and significantly worse than VNS in one case.

### B. Solution Quality, Feasibility and Convergence

Fig. 2 shows how the wake loss depends on the case. It grows with  $N$ , falls as the farm becomes larger, and is higher for the broader wind rose of Data Set II. For  $N \leq 3$ , all methods stay within 0.25% of the wake-free objective, whereas for 15 turbines in the 1000-m farm (Data Set I), the mean wake loss is 1.60% for PSO-VNS, 1.62% for PSO, 1.69% for SSA-VNS and 2.80% for SSA. Over the six largest cases, the mean wake loss of PSO minus that of PSO-VNS ranges from  $-0.18$  to  $+0.41$  percentage points, with PSO-VNS lower in four of them.

PSO-VNS ends with a feasible layout in all runs; the other methods do so in 100.0 (PSO), 99.6 (SSA-VNS), 97.6 (SSA), 96.9 (LX-SSA), 71.4 (DE), 100.0 (VNS) and 100.0% (MS-

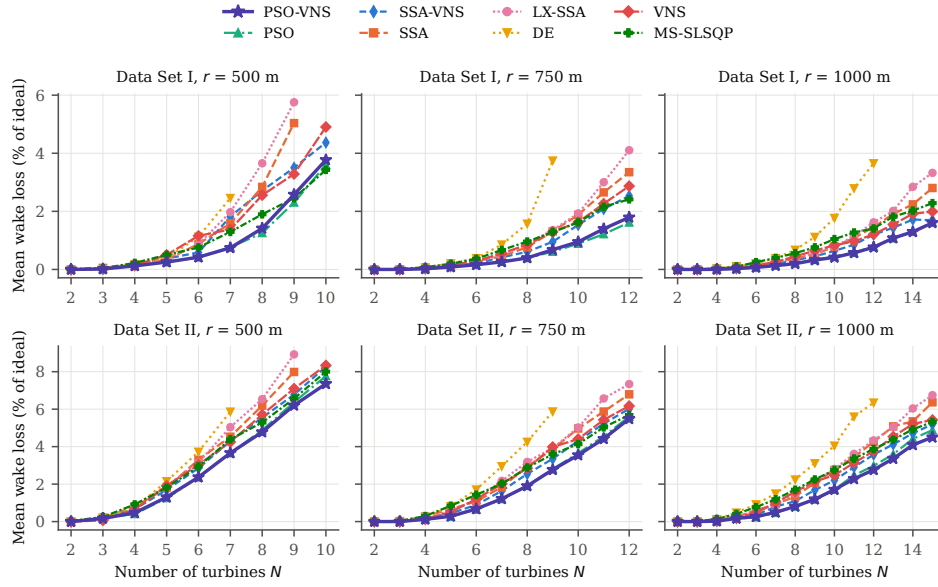


Fig. 2. Mean wake loss (percentage of the wake-free objective) of the feasible runs versus the prescribed number of turbines; a point is omitted when a method has fewer than 15 feasible runs.

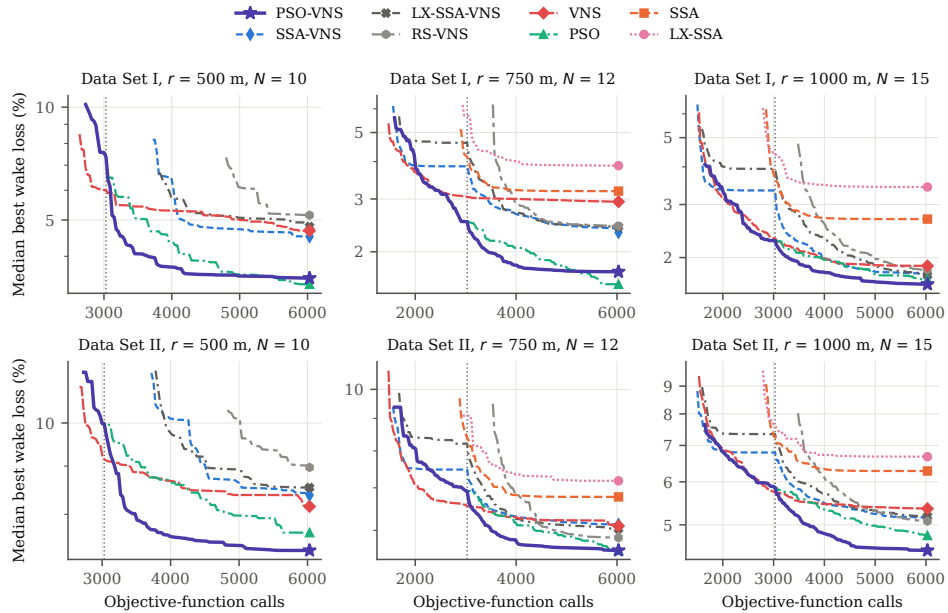


Fig. 3. Convergence of the eight ablation variants (Table III) for the largest turbine count of each farm: median best feasible wake loss over 30 runs versus objective calls. PSO-VNS and PSO coincide up to 3,030 calls (dotted line), where PSO-VNS switches to VNS. The curves of the eight methods of the main comparison are given in Fig. S4 of the supplementary material.

SLSQP) of the runs. The feasibility counts of PSO, SSA, LX-SSA and DE are the same for both wind data sets, because as long as every candidate is infeasible, the penalty dominates  $F_p$ . In the densest cases (500 m,  $N = 10$ ), the global best of the PSO phase is feasible at the switch in only 63.3% of the runs, but all runs are feasible after the VNS phase.

Fig. 3 shows the effect of the second phase directly: PSO-VNS and PSO both keep improving after the switch. Over all feasible runs, the VNS phase removes on average 30.3% (median 22.6%) of the wake loss left at the end of the PSO

phase and turns 26 runs that were still infeasible at the switch into feasible ones, whereas continuing PSO for the same 3,000 calls removes 29.2% (median 20.6%). On average the two uses of the second half therefore differ little, in line with Table II; the advantage of VNS appears in the larger layouts. The VNS phases of SSA-VNS and RS-VNS remove larger shares (36.7% and 57.1%) but start from worse or more often infeasible layouts: at the switch, the best layout is feasible in 98.7% of the PSO-VNS runs, 95.7% of the SSA-VNS runs and 66.7% of the RS-VNS runs. VNS alone ranks below all

TABLE III

ABLATION OVER THE 68 BENCHMARK CASES (30 SEED-PAIRED RUNS, 6,030 CALLS). TOP: AVERAGE RANK AMONG THE EIGHT VARIANTS (FRIEDMAN  $\chi^2_F = 326.8$ , 7 D.F.,  $p = 1.14 \times 10^{-66}$ ) AND FEASIBLE RUNS. BOTTOM: PLANNED CONTRASTS, CASES IN WHICH THE FIRST VARIANT IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (WILCOXON, HOLM-ADJUSTED OVER THE EIGHT CONTRASTS OF EACH CASE,  $\alpha = 0.05$ ), AND MEAN DIFFERENCE OF THE WAKE LOSS  $\Delta\bar{L}$  (PERCENTAGE POINTS; NEGATIVE = FIRST VARIANT BETTER)

| Variant    | Phase 1         | Phase 2 | Avg. rank | Feas. (%) |
|------------|-----------------|---------|-----------|-----------|
| PSO-VNS    | PSO             | VNS     | 1.87      | 100.0     |
| SSA-VNS    | SSA             | VNS     | 3.79      | 99.6      |
| LX-SSA-VNS | LX-SSA          | VNS     | 4.62      | 99.9      |
| RS-VNS     | random sampling | VNS     | 4.46      | 99.4      |
| VNS        | –               | VNS     | 5.32      | 100.0     |
| PSO        | PSO             | –       | 2.10      | 100.0     |
| SSA        | SSA             | –       | 6.56      | 97.6      |
| LX-SSA     | LX-SSA          | –       | 7.29      | 96.9      |

| Contrast               | Isolates                           | W/T/L   | $\Delta\bar{L}$ |
|------------------------|------------------------------------|---------|-----------------|
| PSO-VNS vs. PSO        | VNS phase                          | 8/58/2  | −0.018          |
| PSO-VNS vs. VNS        | swarm start vs. best initial point | 50/17/1 | −0.465          |
| PSO-VNS vs. RS-VNS     | swarm vs. random sampling          | 48/20/0 | −0.399          |
| PSO-VNS vs. SSA-VNS    | PSO vs. SSA as Phase 1             | 47/21/0 | −0.330          |
| PSO-VNS vs. LX-SSA-VNS | PSO vs. LX-SSA as Phase 1          | 48/20/0 | −0.393          |
| SSA-VNS vs. RS-VNS     | SSA vs. random sampling            | 2/66/0  | −0.068          |
| SSA-VNS vs. LX-SSA-VNS | Laplace step in the hybrid         | 3/65/0  | −0.063          |
| LX-SSA vs. SSA         | Laplace step alone                 | 0/61/7  | +0.166          |

hybrids (Table III).

### C. Effect of the PSO Coefficients

The earlier version of this study ran PSO with  $w = 0.7$  and  $c_1 = c_2 = 2$ , which violates (9), on the same 68 cases with the same budget and seeds. With this setting in place of the constriction PSO, PSO had an average rank of 5.12 (fifth of the eight methods), returned a feasible layout in 86.0% of the runs (fewer than 15 feasible runs in 8 cases), and was significantly worse than the constriction PSO in 56 of the 68 cases and never better (Wilcoxon, 30 seed-paired runs), with a mean wake loss 0.61 percentage points higher over the 60 cases in which both have at least 15 feasible runs. The constriction setting thus moves PSO from the fifth to the second rank, ahead of SSA, LX-SSA and SSA-VNS. We draw two conclusions. First, a PSO baseline outside the convergence region understates what PSO achieves on the WFLOP, and comparisons that rely on it can overstate the advantage of a new method; we recommend checking (9), or using the constriction coefficients (8), whenever PSO serves as a baseline. Second, the salp swarm methods, including our own LX-SSA, do not outperform a correctly configured PSO on this benchmark, which is why PSO rather than SSA forms the first phase of the proposed method.

### D. Ablation

To see where the improvement of PSO-VNS comes from, we compare it with variants that each remove or replace one ingredient (Table III and Fig. 3). The Friedman test over the eight variants gives  $\chi^2_F = 326.8$  (7 d.f.,  $p = 1.1 \times 10^{-66}$ ), and the variants rank PSO-VNS (1.87), PSO (2.10), SSA-VNS (3.79), RS-VNS (4.46), LX-SSA-VNS (4.62), VNS (5.32), SSA (6.56) and LX-SSA (7.29).

TABLE IV

BUDGET SPLIT OF PSO-VNS (25%, 50% AND 75% OF THE 6,030 CALLS FOR PSO) ON 12 CASES (MIDDLE AND LARGEST  $N$  OF EACH FARM AND DATA SET, 30 SEEDS): MEAN WAKE LOSS, AVERAGE RANK AMONG THE THREE SPLITS, AND CASES IN WHICH THE DEFAULT 50% SPLIT IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (WILCOXON, HOLM-ADJUSTED)

| PSO share $\omega$ | Mean loss (%) | Avg. rank | 50% vs. $\omega$ (W/T/L) |
|--------------------|---------------|-----------|--------------------------|
| 25%                | 2.796         | 2.92      | 6/6/0                    |
| 50%                | 2.643         | 2.00      | –                        |
| 75%                | 2.543         | 1.08      | 0/7/5                    |

*VNS phase.* Since PSO-VNS and PSO share their first 3,030 calls, this contrast measures only the use of the second half of the budget. Switching to VNS instead of continuing the swarm gives a small gain overall that grows with  $N$  (Section VI-A; 8/58/2 under the Holm adjustment over the contrasts of Table III): a convergent PSO keeps improving in the second half on small layouts, whereas on larger layouts the single-turbine moves of the compass search appear to pay off.

*Swarm start.* Starting VNS from the PSO global best rather than from the best initial point is significantly better in 50 cases and worse in 1 (mean wake-loss difference  $-0.465$  percentage points). The random-multistart control RS-VNS spends the same 3,015 calls of Phase 1 on independent random layouts; PSO-VNS is significantly better than RS-VNS in 48 cases and never worse ( $-0.399$  percentage points), so the PSO phase adds value beyond a good random start. For SSA the answer is negative: SSA-VNS and RS-VNS are statistically tied in 66 of the 68 cases (2/66/0).

*Choice of the Phase-1 swarm.* With the same VNS phase, PSO-VNS is 47/21/0 (W/T/L) against SSA-VNS and 48/20/0 against LX-SSA-VNS, so a convergent PSO is the better first phase. At the switch, its global best is feasible more often (98.7% of the runs vs. 95.7% and 94.1%) and, when feasible, has a lower wake loss (1.63% vs. 2.01% and 2.11%), so VNS starts from a better layout.

*Laplace step.* The Laplace step of LX-SSA gives no gain: without the VNS phase, LX-SSA is 0/61/7 against SSA, and inside the hybrid, SSA-VNS is 3/65/0 against LX-SSA-VNS (mean wake-loss difference  $-0.063$  percentage points).

### E. Budget Split

Table IV shows the effect of the split  $\omega$  on the twelve cases with the middle and the largest  $N$  of each farm (per-case values in Table S12 of the supplementary material). The default  $\omega = 0.5$ , fixed before the experiments, is not the best choice on these cases: with  $\omega = 0.75$ , the mean wake loss is lower than with  $\omega = 0.5$  in eleven of the twelve cases, significantly in five, and 0.10 percentage points lower on average, whereas  $\omega = 0.25$  is significantly worse than the default in six cases. A longer PSO phase therefore pays off on this benchmark, consistent with the observation that PSO keeps improving after 3,030 calls. We keep the pre-specified  $\omega = 0.5$  for all other results, so that no parameter of PSO-VNS is tuned on the test cases; the main comparison therefore tends

TABLE V  
SIX LARGEST BENCHMARK CASES: MEAN WAKE LOSS (%) AND FEASIBLE RUNS (%) AT 6,030 CALLS WITH RANDOM AND FEASIBILITY-PRESERVING INITIALIZATION, AND AVERAGE RANK AT 6,030, 30,030 AND 120,030 CALLS (RANDOM INITIALIZATION; 30 SEEDS); N/R: NOT RUN

| Method     | Random init. |       | Feasible init. |       | Avg. rank |        |         |
|------------|--------------|-------|----------------|-------|-----------|--------|---------|
|            | Loss         | Feas. | Loss           | Feas. | 6,030     | 30,030 | 120,030 |
| PSO-VNS    | 4.084        | 100.0 | 3.938          | 100.0 | 1.50      | 1.17   | 2.17    |
| SSA-VNS    | 4.678        | 95.0  | 3.955          | 100.0 | 4.33      | 4.17   | 4.17    |
| LX-SSA-VNS | 4.802        | 98.9  | 3.952          | 100.0 | 5.17      | 5.67   | 5.83    |
| RS-VNS     | 4.830        | 93.3  | n/r            | n/r   | 4.67      | 6.33   | 4.67    |
| VNS        | 4.950        | 100.0 | 3.810          | 100.0 | 6.50      | 4.83   | 3.17    |
| PSO        | 4.181        | 100.0 | 5.549          | 100.0 | 1.83      | 2.50   | 5.83    |
| SSA        | 4.828        | 80.0  | 4.947          | 100.0 | 8.33      | 8.67   | 8.83    |
| LX-SSA     | 5.379        | 77.8  | 5.048          | 100.0 | 8.67      | 9.83   | 10.00   |
| DE         | —            | 1.1   | 5.549          | 100.0 | 10.00     | 5.17   | 3.33    |
| MS-SLSQP   | 4.522        | 99.4  | 4.294          | 100.0 | 4.00      | 6.67   | 7.00    |

to understate PSO-VNS slightly, and we recommend  $\omega = 0.75$  as the starting point for applications.

## VII. RESULTS BEYOND THE BENCHMARK

### A. Feasibility-Preserving Initialization and Budget Scaling

Table V summarizes the six largest benchmark cases. With feasibility-preserving initialization, the share of feasible runs rises for SSA-VNS (95.0% to 100.0%), SSA (80.0% to 100.0%), LX-SSA (77.8% to 100.0%) and DE (1.1% to 100.0%); the mean wake loss of PSO-VNS changes from 4.08% to 3.94%, and VNS has the lowest mean wake loss (per-case values in Table S15). PSO and DE, in contrast, never improve on the best initial layout in these runs: since the turbine labels are arbitrary, velocity and difference moves that combine two feasible layouts almost always make turbines overlap, the feasibility-first comparison rejects them, and the global-best particle of PSO does not move. The single-coordinate moves of VNS are not affected, so a feasible start favors the VNS-based methods (PSO-VNS ranks 2.83 on average, behind VNS alone), whereas the penalty-guided search from random starts favors PSO.

### B. Horns Rev 1 16-Turbine Block

Table VI gives the results for the 16-turbine block; the installed layout, a regular 7D grid, reaches 139.51 GWh/yr (wake loss 6.53%). PSO-VNS finds a feasible layout in all 30 runs, with a mean AEP of 138.73 GWh/yr (SD 0.83) at 6,030 calls, the highest of all methods and significantly higher than that of every other method ( $p_{\text{Holm}} \leq 6.7 \times 10^{-4}$ ). Its best run reaches 140.39 GWh/yr and exceeds the installed layout, as do 6 of its 30 runs; of the other methods, only PSO (one run) does so. From random starts, PSO finds a feasible layout inside the parallelogram in only 19/30 runs; with feasibility-preserving initialization it does so in 30/30 runs, and at 30,030 and 120,030 calls in 28/30 and 10/10 runs. With larger budgets, the mean AEP loss of PSO-VNS falls from 7.05% to 6.46% (30,030 calls) and 6.45% (120,030 calls). The best layouts are shown in Fig. 5.

TABLE VI  
HORNS REV 1 16-TURBINE BLOCK (WAKE-FREE AEP 149.26 GWh/yr): MEAN AEP (GWh/yr) OVER THE FEASIBLE RUNS OF 30 SEEDS AT 6,030 CALLS, FEASIBLE RUNS, HOLM-ADJUSTED WILCOXON  $p$  OF PSO-VNS VS. EACH METHOD, AND MEAN AEP LOSS (%) AT 6,030 CALLS WITH RANDOM (R) AND FEASIBILITY-PRESERVING (F) INITIALIZATION AND AT 30,030 AND 120,030 CALLS (R; 10 SEEDS AT 120,030); SUPERScript: FEASIBLE RUNS WHEN NOT ALL RUNS ARE FEASIBLE

| Layout / method  | 6,030 calls, R |       |                      | Loss (%)           |      |                    |      |
|------------------|----------------|-------|----------------------|--------------------|------|--------------------|------|
|                  | AEP            | Feas. | $p_{\text{Holm}}$    | 6k R               | 6k F | 30k                | 120k |
| Installed layout | 139.51         | —     | —                    | 6.53               | —    | —                  | —    |
| PSO-VNS          | 138.73         | 30/30 | —                    | 7.05               | 6.96 | 6.46               | 6.45 |
| SSA-VNS          | 137.98         | 30/30 | $6.7 \times 10^{-4}$ | 7.56               | 6.99 | 7.16               | 6.48 |
| LX-SSA-VNS       | 137.77         | 30/30 | $6.3 \times 10^{-4}$ | 7.70               | 7.06 | 7.06               | 6.51 |
| RS-VNS           | 137.96         | 28/30 | $2.2 \times 10^{-4}$ | 7.57 <sup>28</sup> | n/r  | 6.98               | 6.24 |
| VNS              | 137.67         | 30/30 | $2.2 \times 10^{-4}$ | 7.77               | 6.88 | 6.72               | 6.12 |
| PSO              | 138.06         | 19/30 | $1.4 \times 10^{-5}$ | 7.51 <sup>19</sup> | 8.39 | 6.82 <sup>28</sup> | 6.66 |
| SSA              | 136.66         | 30/30 | $2.8 \times 10^{-7}$ | 8.44               | 7.89 | 8.14               | 7.61 |
| LX-SSA           | 136.46         | 28/30 | $1.5 \times 10^{-7}$ | 8.58 <sup>28</sup> | 7.96 | 8.48               | 8.36 |
| DE               | —              | 0/30  | $1.7 \times 10^{-8}$ | —                  | 8.39 | 7.28 <sup>8</sup>  | 6.34 |
| MS-SLSQP         | 136.80         | 30/30 | $1.5 \times 10^{-7}$ | 8.35               | 8.06 | 7.79               | 7.26 |

TABLE VII  
IEA37 CASE STUDY 1, 16- AND 36-TURBINE SCENARIOS: AEP IN MWh OF THE BEST RUN OF EACH METHOD AT 6,030 AND 30,030 CALLS (30 SEEDS), COMPARED WITH THE EXAMPLE LAYOUT AND THE BEST FEASIBLE PUBLISHED LAYOUT [20], [36]

| Method / source                      | 16 turbines ( $r = 1300$ m) |           | 36 turbines ( $r = 2000$ m) |           |
|--------------------------------------|-----------------------------|-----------|-----------------------------|-----------|
|                                      | 6,030                       | 30,030    | 6,030                       | 30,030    |
| Example layout                       | 366,941.6                   |           | 737,883.1                   |           |
| Best feasible published <sup>a</sup> | 418,924.4                   |           | 863,676.3                   |           |
| PSO-VNS                              | 402,782.4                   | 413,331.9 | 787,708.9                   | 827,385.0 |
| PSO                                  | 403,753.1                   | 412,702.4 | 787,988.7                   | 830,256.4 |
| SSA-VNS                              | 403,251.3                   | 406,924.6 | 766,803.3                   | 819,930.5 |
| SSA                                  | 395,150.2                   | 401,459.3 | 750,707.4                   | 802,538.2 |
| LX-SSA                               | 393,284.6                   | 398,770.0 | 746,301.3                   | 789,311.3 |
| DE                                   | 357,391.3                   | 397,402.7 | —                           | —         |
| VNS                                  | 400,169.3                   | 404,076.0 | 785,791.3                   | 824,230.2 |
| MS-SLSQP                             | 405,433.8                   | 410,349.5 | 748,571.1                   | 833,365.0 |

<sup>a</sup>Participant 4 in both scenarios (SNOPT with wake expansion continuation [TBD: verify against Baker et al. (2019)]). A higher submitted AEP places turbines up to 3.5 m outside the boundary and is not counted.

### C. IEA37 Case Study 1

Table VII and Fig. 5 place our methods in the context of IEA37 Case Study 1, in which the participants chose their own optimizers and budgets [20]. The best feasible submitted layouts reach 418,924.4 MWh for 16 turbines and 863,676.3 MWh for 36 turbines. At 30,030 calls, the best PSO-VNS layout would rank third among the 9 feasible submitted 16-turbine layouts and seventh among the 8 feasible 36-turbine layouts; it differs from the best feasible published layout by  $-1.33\%$  and  $-4.20\%$  ( $-3.85\%$  and  $-8.80\%$  at 6,030 calls), and the mean over the 30 runs by  $-3.90\%$  and  $-6.51\%$ . Since the participants' optimizers and budgets differ and are not reported in a comparable form, this comparison is indicative only.

### D. Published Kusiak–Song Results and Computational Cost

A comparison with the published results of the Kusiak–Song benchmark [3]–[6] is given in Section S-V of the supplementary material [TBD: pending author verification of the PDFs]. At equal numbers of objective calls, all methods

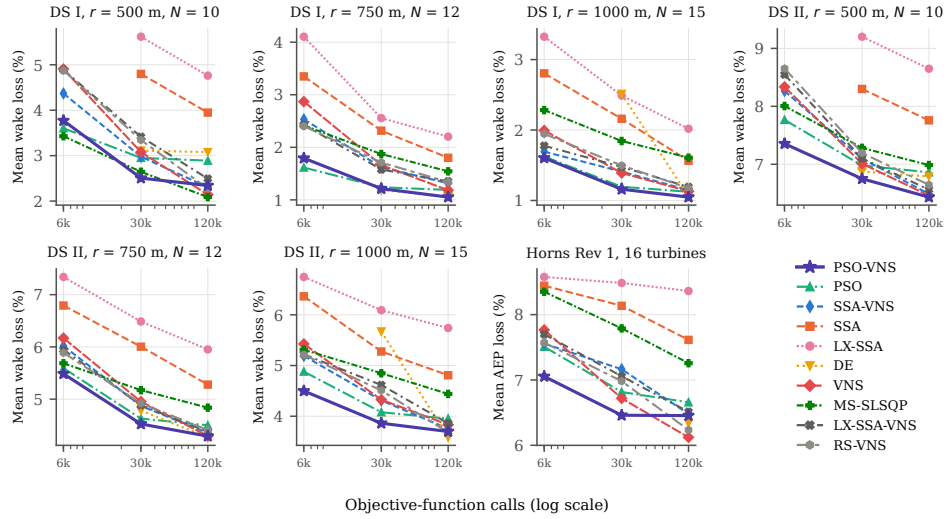


Fig. 4. Budget scaling: mean wake loss of the six largest benchmark cases and mean AEP loss of the Horns Rev 1 block (feasible runs; methods with at least half of their runs feasible) versus the budget (6,030, 30,030 and 120,030 calls, random initialization; 10 seeds for Horns Rev 1 at 120,030 calls).

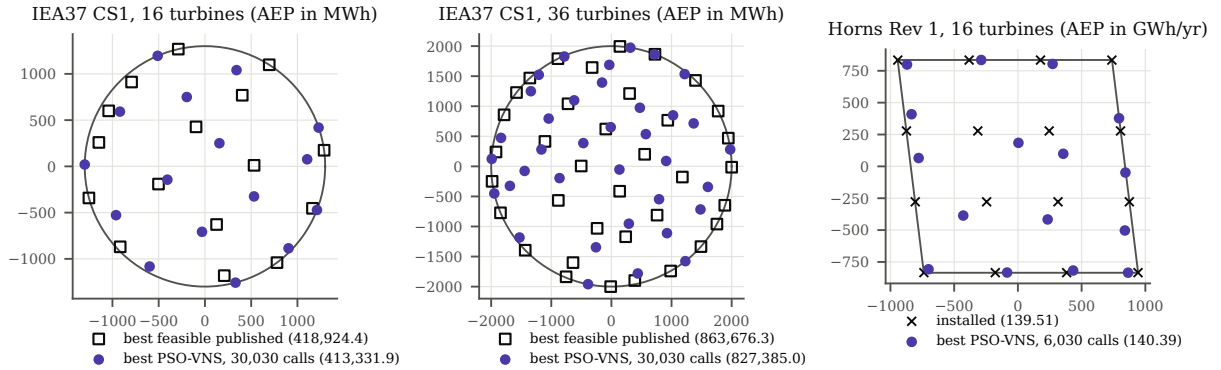


Fig. 5. Best layouts. Left and middle: IEA37 Case Study 1, 16 and 36 turbines, best PSO-VNS layout at the largest budget and best feasible published layout. Right: Horns Rev 1 16-turbine block, installed layout and best PSO-VNS layout at 6,030 calls.

need a similar time (0.24–1.82 ms per call on the machines that produced the respective runs; Table S11); one 6,030-call run of PSO-VNS takes 1.5–3.2 s, depending on  $N$ .

## VIII. SENSITIVITY TO THE MODELING ASSUMPTIONS

### A. Power Curve and Wake Model

We re-evaluated every feasible final layout of the main comparison, without re-optimization, with 1) a cubic power curve [40],  $f_c(s) = P_{\text{rated}}(s^3 - s_{\text{cut-in}}^3)/(s_{\text{rated}}^3 - s_{\text{cut-in}}^3)$  between the cut-in and rated speeds, 2) the same curve with a 25 m/s cut-out, and 3) the Gaussian wake model of Bastankhah and Porté-Agel [41] with growth rate  $K_G = 0.04$  and the same superposition and Weibull scaling as the benchmark (Section S-I.B of the supplementary material).

Table VIII shows two kinds of sensitivity. First, the absolute energy depends strongly on the model: the cubic curve gives 17.7% (Data Set I) and 36.9% (Data Set II) less expected power than the linear benchmark curve, so the energy values in this paper are benchmark values and not AEP predictions for a specific turbine. Second, the ranking of the algorithms is

TABLE VIII  
ROBUSTNESS TO THE BENCHMARK MODEL: ALL FEASIBLE FINAL LAYOUTS RE-EVALUATED (NOT RE-OPTIMIZED).  $\Delta$ : MEAN RELATIVE CHANGE OF THE OBJECTIVE (%); RANK: AVERAGE RANK OF PSO-VNS OVER THE 68 CASES; BEST: BEST-RANKED METHOD;  $\bar{\tau}$ : MEAN KENDALL CORRELATION BETWEEN THE BENCHMARK AND ALTERNATIVE ORDERINGS WITHIN A CASE; SAME: CASES (%) WITH UNCHANGED BEST METHOD. THE RANKS OF ALL METHODS ARE GIVEN IN TABLE S20

| Model                 | $\Delta$ (%) |       |      | Rank    | Best | $\bar{\tau}$ | Same |
|-----------------------|--------------|-------|------|---------|------|--------------|------|
|                       | DS I         | DS II |      |         |      |              |      |
| Benchmark             | +0.0         | +0.0  | 1.82 | PSO-VNS | –    | –            | –    |
| Cubic curve           | -17.7        | -36.9 | 1.82 | PSO-VNS | 0.95 | 94           | 94   |
| Cubic, 25 m/s cut-out | -21.5        | -37.0 | 1.80 | PSO-VNS | 0.95 | 93           | 93   |
| Gaussian wake         | -0.3         | -0.4  | 1.97 | PSO     | 0.66 | 54           | 54   |

stable under the cubic curve ( $\bar{\tau} = 0.95$ ; same best method in 94% of the cases) and less stable under the Gaussian model ( $\bar{\tau} = 0.66$ ; same best method in 54% of the cases), because layouts optimized for the sharp Jensen wake cone are judged differently by a smooth Gaussian deficit. Under the Gaussian model, PSO-VNS ranks second (1.97), just behind

PSO (1.91), and both remain ahead of all other methods; the small advantage of the VNS phase is thus specific to the model it optimizes.

### B. Minimum Spacing and Boundary Handling

The 4D spacing is inherited from the benchmark literature. A multi-start packing search constructed feasible layouts with up to 13, 26 and 42 turbines at 4D, 8, 19 and 29 at 5D, and 7, 13 and 21 at 6D for  $r = 500, 750$  and  $1000$  m (constructive lower bounds, Table S8), so the 500-m cases with  $N \geq 9$  (at 5D) or  $N \geq 8$  (at 6D) may be infeasible. The optimized layouts often lie on the 4D constraint: of the feasible final layouts of the eight methods, 49% also satisfy 5D and 36% satisfy 6D (4.1% and 0.1% for  $N = 11$ –15), so they cannot be transferred to sites with larger spacing without re-optimization; a re-optimization of six cases at 5D and 6D with SSA and LX-SSA is reported in Table S7 of the supplementary material.

The boundary rule also matters: an otherwise identical SSA that projects every turbine radially onto the circle instead of clipping the coordinates attains higher mean benchmark objectives than the original SSA in six representative cases at 3,030 calls, by 26 to 649 units (Mann–Whitney  $p < 10^{-4}$  in every case). We therefore keep the rule of Section II-C fixed for all methods.

## IX. LIMITATIONS

The components of PSO-VNS are standard, and most of its advantage over the other methods comes from the convergent PSO phase: its gain over PSO alone is small and not significant over all 68 cases ( $p = 0.30$ ), with significant differences in 18 cases that mostly favor PSO-VNS and are more frequent for  $N \geq 10$ . The budget study covers up to 120,030 calls but only the six largest benchmark cases and the Horns Rev 1 block, and the largest layouts have 36 turbines; much larger farms are outside our scope. The PSO coefficients are the standard constriction values, and the VNS settings and  $\omega = 0.5$  are untuned defaults; the split study indicates that a longer PSO phase would improve PSO-VNS. Our statement on the setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  rests on the convergence theory and our own benchmark; we did not re-run the comparisons of other studies. The absolute energy depends strongly on the Jensen model, the linear power curve, the 4D spacing and the boundary rule, the order of PSO-VNS and PSO is reversed under the Gaussian model, and we did not re-optimize with the alternative models. Finally,  $N$  is prescribed, and terrain, mixed turbine types, cables, grid connection, noise, environmental constraints and economic objectives are outside the scope of this study.

## X. CONCLUSION

This work presents PSO-VNS, a two-phase algorithm for the continuous wind farm layout problem in which PSO with constriction coefficients explores the layout space for the first half of the budget and the basic VNS then refines the global best layout. We evaluated it on 68 benchmark cases at equal budgets, in an ablation, with feasibility-preserving

initialization and larger budgets, on a Horns Rev 1 block and on IEA37 Case Study 1.

PSO-VNS returned a feasible layout in every run, obtained the best average rank (1.82), and was at least as good as a correctly configured PSO, with its gain concentrated in the larger cases (9 wins and 2 losses for  $N \geq 10$ ). It clearly outperformed SSA-VNS (49 wins, no loss), SSA, LX-SSA, DE, VNS and MS-SLSQP. The ablation showed that PSO is a better first phase than SSA or LX-SSA; that the PSO phase gives VNS a better start than the best initial layout and than the best of an equal number of random layouts, whereas the SSA phase is statistically tied with random sampling in nearly all cases; and that the Laplace step of our earlier LX-SSA did not improve the results, either alone or inside the hybrid. PSO configured inside its convergence region outperformed the salp swarm methods, whereas the common setting  $w = 0.7$ ,  $c_1 = c_2 = 2$  violates the convergence condition and understates PSO; PSO baselines in WFLOP comparisons should be checked against this condition. On the Horns Rev 1 block, the best PSO-VNS layout exceeded the energy production of the installed layout. With feasibility-preserving initialization, the share of feasible runs rises for SSA-VNS (95.0% to 100.0%), SSA (80.0% to 100.0%), LX-SSA (77.8% to 100.0%) and DE (1.1% to 100.0%); with up to 20 times the budget, PSO-VNS keeps the best average rank at all three budgets; and on IEA37 Case Study 1, the best PSO-VNS layouts reach 98.7% (16 turbines) and 95.8% (36 turbines) of the AEP of the best feasible published layouts. The ranking of the methods was stable under a cubic power curve, whereas under a Gaussian wake model PSO ranked slightly ahead of PSO-VNS. Finally, giving PSO 75% instead of 50% of the budget improved PSO-VNS on most cases of the split study.

In future work, we will study a switch between the two phases that is triggered by the stagnation of the swarm, investigate which properties of the swarm phase make it a good starting point for VNS, and extend the approach to larger farms and higher-fidelity wake and power models.

## ACKNOWLEDGMENT

[TBD: acknowledgment, or delete this section]

## DATA AVAILABILITY

The source code, the per-run results (including the final coordinates and convergence curves of all runs) and the scripts that generate all tables and figures are openly available at [TBD: Zenodo DOI]. The supplementary material contains the model details, the pseudocode of SSA and LX-SSA, the wind data, the per-case results of all studies, the coordinates of the best layouts and further figures.

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